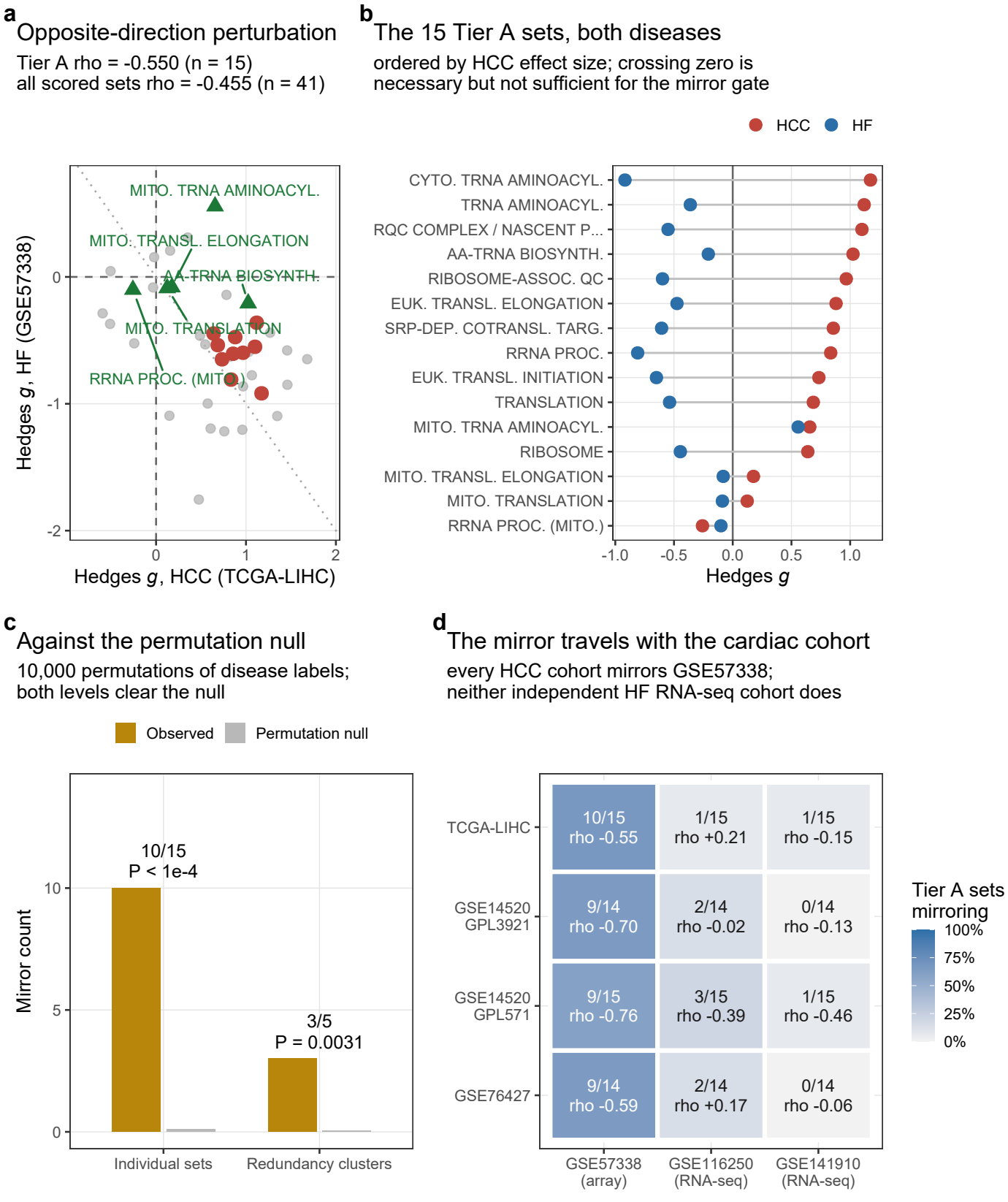
a
Opposite-direction perturbation
Tier A rho = -0.550 (n = 15)
all scored sets rho = -0.455 (n = 41)
MITO. TRNA AMINOACYL.
MITO. TRANSL. ELONGATION
AA-TRNA BIOSYNTH.
MITO. TRANSLATION
RRNA PROC. (MITO.)
Hedges g, HF (GSE57338)
Hedges g, HCC (TCGA-LIHC)
b
The 15 Tier A sets, both diseases
ordered by HCC effect size; crossing zero is
necessary but not sufficient for the mirror gate
HCC
HF
CYTO. TRNA AMINOACYL.
TRNA AMINOACYL.
RQC COMPLEX / NASCENT P...
AA-TRNA BIOSYNTH.
RIBOSOME-ASSOC. QC
EUK. TRANSL. ELONGATION
SRP-DEP. COTRANSL. TARG.
RRNA PROC.
EUK. TRANSL. INITIATION
TRANSLATION
MITO. TRNA AMINOACYL.
RIBOSOME
MITO. TRANSL. ELONGATION
MITO. TRANSLATION
RRNA PROC. (MITO.)
Hedges g
c
Against the permutation null
10,000 permutations of disease labels;
both levels clear the null
Observed
Permutation null
10/15
P < 1e-4
3/5
P = 0.0031
Mirror count
Individual sets
Redundancy clusters
d
The mirror travels with the cardiac cohort
every HCC cohort mirrors GSE57338;
neither independent HF RNA-seq cohort does
TCGA-LIHC
GSE14520 GPL3921
GSE14520 GPL571
GSE76427
10/15 rho -0.55
1/15 rho +0.21
1/15 rho -0.15
9/14 rho -0.70
2/14 rho -0.02
0/14 rho -0.13
9/15 rho -0.76
3/15 rho -0.39
1/15 rho -0.46
9/14 rho -0.59
2/14 rho +0.17
0/14 rho -0.06
GSE57338 (array)
GSE116250 (RNA-seq)
GSE141910 (RNA-seq)
Tier A sets mirroring
100%
75%
50%
25%
0%